

NANOSCIENCE And NANOTECHNOLOGY In India: What's Happening



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Advancements in science and technology have led to the emergence of nanoscience in India. This has further led to the development of various specialised courses for nanoscience and nanotechnology. Universities and institutions in India have started bachelor's, master's and research programmes in nanoscience and, subsequently, nanotechnology as an extended branch of basic sciences in India. Most nanoscience courses in India have emerged out of Physics, Chemistry, Life Science and other engineering departments of various institutions.

Introduction of nanoscience and nanotechnology started as an elective course. Then, with permission from apex bodies, institutions started full-fledged courses

(mostly two-year ones). A few institutions have also started research programmes in interdisciplinary areas of nanoscience and nanotechnology. In some universities, these have come out as independent departments or centres, while in others as part of core departments.

Infrastructure facilities were not good 10 years ago, but now in many institutions, these are self-sufficient to run nanoscience and nanotechnology programmes. Teachers have been inducted from various basic science and engineering departments to contribute in their areas of expertise. In a few institutions, permanent teaching faculties have been hired to run the programmes under schemes such as special or innovation centres, or centres of excellence (CoEs).



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TABLE I

NATIONAL LABS/INSTITUTIONS/UNIVERSITIES INVOLVED IN NANOSCIENCE AND NANOTECHNOLOGY ACTIVITIES

Labs	Institutions	Universities
• DST: ARCI, IACS, SBNBCBS, CeNS, INST, ARI and JNCASR • DRDO: SSPL, DMSRDE, DMRL, NMRL, HEMRL, NPOL, ACEM and DLJ • CSIR: IICT, NPL, NCL, INST, NIIST, CLRI, CGCRI, NIIST, IMMT, AMPRI, IITR, CSIO, CEERI, CECRI, CFTRI, CMERI, IMT, IIIM, NAL and NML • ISRO: VSSC and Semiconductor Complex Mohali • UGC: UGC-DAE Consortium (Kalpakkam, Mumbai, Kolkata, Indore), IUAC, CGC-Anna University Campus • DAE: BARC, IGCAR, RRCAT, IOP, FCIPT-IPR, TIFR, SINP and NCBS • MeitY: C-MET (Hyderabad, Pune, Thrissur) • DBT: ILS, CIAB and NABI • ICAR: CIRCOT, CIFE, CIFA and NIASM • MNRE: NISE and SSS-NIRE • MoFPI: IIFPT and NIFTEM • ICMR Labs: Regulations on nanotechnology and handling of nano-materials • Government-recognised Semiconductor Technology and Applied Research Centre (STARC)-formerly SITAR, Bengaluru • Government-recognised Gallium Arsenide Enabling Technology Centre (GAETEC), Hyderabad	• IISc Bangalore • NITs (Rourkela, Trichy, Warangal, Calicut, Bhopal, Nagpur, Jaipur, Allahabad and a few newly established NITs) • IITs (Kharagpur, Kanpur, Mumbai, Madras, Roorkee, Delhi, Guwahati and a few newly established IITs like Indore, Ropar, Gandhinagar, Patna, Bhubaneswar, etc) • IISERs (Kolkata, Pune, Mohali, Bhopal, Trivendrum and Berhampur) • IEST (formerly BESU/BEC), Shibpur, West Bengal	• Central universities like Jawaharlal Nehru University, Delhi University, Pondicherry University, Allahabad University, Tezpur University, University of Hyderabad, CU-Jharkhand • State universities like Jadavpur University, Calcutta University, Madras University, Bangalore University, Mysore, Karnataka University, Bharathiar University, Bharathidasan University, Alagappa University, Osmania University, Madurai Kamraj University, Anna University, Utkal University, Sambalpur University, North-Orissa University, Mumbai University, Shivaji University, Andhra University, TERI, SV University, Punjab University, University of Rajasthan • Leading private universities like SASTRA Deemed University, VIT Deemed University, SRM Deemed University, Sathyabama Deemed University, Amity University, Amrita Deemed University, Karunya Deemed University, Periyar Maniammai Deemed University, Hindustan Deemed University, Annamalai University, BIT Mesra, BITS Pillani

This article discusses the scope of nanoscience and nanotechnology along with curricular courses, initiated by various Indian institutions